

Lab 3:

Component Modelling & Architectural Pattern Selection

Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice

The Layered Architecture is selected because the coffee kiosk has a clear separation between the user interface, business logic, and data storage. This makes the system easier to understand and maintain.

Reason 1

The architecture separates the touch-screen interface from the Order Manager, allowing customer interactions and order processing to be handled independently.

Reason 2

The architecture separates the Database Component and hardware-related components such as the Receipt Printer and Payment Service, making the system more organized and easier to maintain.

Security Advantage

The separation of components helps isolate sensitive payment processing from the user interface and other parts of the system. This can reduce unnecessary access to payment-related functionality.

Performance Benefit

Separating the components allows each part of the system to perform its specific task efficiently.

For example, the Order Manager can process orders while the Database handles menu and pricing information.